EXPERIMENT 10

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Experiment 10 – Trigger for Automatic Primary Key

Experiment 10: Design and Implementation of a Trigger for Automatic Primary Key Generation

Aim

To design a trigger that automatically implements the functionality of a primary key, ensuring unique identification of records without manual intervention.

Objective

- To understand the concept and working of database triggers
- To implement automatic primary key generation using triggers
- To eliminate manual intervention in key assignment
- To ensure uniqueness and data integrity
- To apply trigger-based automation in enterprise systems

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Client Tools:

- Oracle SQL Developer
- pgAdmin

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Problem Statement

In enterprise-level applications, manually assigning primary keys can lead to duplication and errors. To maintain data integrity and ensure uniqueness, a trigger-based mechanism is required that automatically generates a primary key before inserting records into a table.

Practical / Experiment Steps

- Create a table requiring automatic primary key generation
- Create a sequence to generate unique values
- Design a trigger that activates before insert
- Assign the next sequence value to the primary key
- Insert multiple records to test the trigger
- Verify uniqueness of generated keys

Procedure

- Open Oracle SQL Developer and connect to the database
- Create the required table
- Create a sequence for primary key generation
- Create a trigger using BEFORE INSERT event
- Compile the trigger successfully
- Insert records without specifying the primary key
- Display the table contents

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Input / Output Details

Input

Table: employee (emp_id, name, salary)

SQL Implementation Overview

- **Table creation**
- **Sequence creation**
- **Trigger creation**
- **Data insertion**
- **Result verification**

Output

After executing the program, each record is automatically assigned a unique primary key without manual input.

Example Output:
1 Aman 40000
2 Neha 50000
3 Rahul 45000
4 Simran 55000

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the working of database triggers
- Implement automatic primary key generation
- Ensure data integrity without manual intervention
- Apply trigger-based solutions in enterprise applications
- Develop efficient and reliable database systems